

strategies are usually expressed in terms of percentage of market volume (POV rate) or in terms of trading time.

*Alpha Capture.* Managers determine the expected profit level of an investment idea based on the stocks projected alpha and the corresponding trading cost. This helps to maximize expected ex-post stock return (e.g., returns after incurring trading cost). Managers then select stocks based on ex-post return.

*Investment Capacity.* Determine how many shares of a stock can be transacted before the trading cost erodes the expected stock return beyond a specified level. For example, suppose that a manager has determined that an investment strategy is expected to achieve an incremental return of 3% over its benchmark. The manager turns to TCA to determine how many shares can be purchased with a trading cost equal to the incremental return of 3%. After this point, the manager is better off investing in their next most attractive investment idea.

*Portfolio Optimization.* Portfolio optimization techniques are being developed to provide managers with the “optimal” weightings and the underlying transaction strategy to achieve those positions. These optimizations incorporate expected returns, volatility, correlation, and market impact to determine the optimal mix of stock. Market impact cost is determined from the underlying market impact model parameters. And, the resulting execution strategy is the “best execution” strategy that provides exact consistency between investing and trading decisions. The optimization technique will take advantage of any synergies resulting from diversification or market hedging opportunities. Portfolio optimization with transaction cost analysis has become one of leading areas of research for portfolio managers and is discussed in Chapter 10, Portfolio Construction.

*Back Testing.* Managers use market impact back-testing series to test investment ideas and determine if those ideas will be profitable in different market conditions. All too often, however, managers find a strategy works well in the back-testing environment but once the strategy goes live it does not provide the expected level of return due to implementation costs. Some of the more forward thinking managers have begun incorporating historical trading costs into their back-testing scenarios. The biggest issue we have encountered here is that while there are participants providing historical costs these are based on the actual market structure and actual cost of trading at that specific point in time. This could result in dramatically overstating the true cost of trading (such as in the early to mid-1990s) when stocks were quoted in 1/8ths (well really odd-eighths or quarters, see Christie and Stoll 1994). Overstating the true costs in a back-testing environment could